Linear Algebra for AI/ML

A Concise, Analogy-First Guide

Learning Framework: For every topic we follow the same four steps:

1. Real-Life Analogy
2. Mathematics
3. Python
4. Machine Learning Connection

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1. Scalar

Real-Life Analogy

The bill at a restaurant is Rs. 500. Your age is 25. Today's temperature is 30°C. Each is just *one* number standing alone — that is a scalar.

Mathematics

$$x = 5 \qquad \text{or} \qquad x = 3.14$$

Python

```
age = 25
salary = 50000
temperature = 30.5
```

ML Connection

The final prediction of a regression model (e.g. predicted house price = Rs. 45,00,000) is a scalar.

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2. Vector

Real-Life Analogy

A student's profile card: Study Hours = 5, Attendance = 90, Assignments = 8. Bundling these three numbers together as one object gives a vector.

Mathematics

$$\mathbf{x} = \begin{bmatrix} 5 \\ 90 \\ 8 \end{bmatrix}$$

Python

```
import numpy as np
student = np.array([5, 90, 8])
```

ML Connection

One row of any dataset (one student, one customer, one house) is stored as a vector of features.

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3. Vector Addition

Real-Life Analogy

Monday you walk 2 km east, 1 km north. Tuesday you walk 3 km east, 4 km north. Your total displacement is found by adding the two trips component-by-component.

Mathematics

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

Python

```
a = np.array([2, 1])
b = np.array([3, 4])
print(a + b)    # [5 5]
```

ML Connection

Adding the bias vector to weighted inputs, accumulating gradients, and combining embeddings all rely on vector addition.

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4. Matrix

Real-Life Analogy

A classroom register — many students (rows), several columns of information (hours, attendance, assignments). A matrix is simply a mathematical table.

Mathematics

$$A = \begin{bmatrix} 5 & 90 & 8 \\ 3 & 70 & 6 \\ 7 & 95 & 10 \end{bmatrix}$$

Python

```
students = np.array([
[5, 90, 8],
[3, 70, 6],
[7, 95, 10]
])
```

ML Connection

Nearly every dataset is a matrix: rows are samples, columns are features.

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5. Shape

Real-Life Analogy

Describing an Excel sheet as "3 rows by 4 columns" — that is its shape.

Mathematics

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, shape $= (2, 3)$.

Python

```
A.shape    # (2, 3)
```

ML Connection

A dataset of 1000 students and 5 features has shape `(1000, 5)` — the first thing every data scientist checks.

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6. Transpose

Real-Life Analogy

Rotating an attendance sheet so that the column "Marks" becomes a row across the top. Rows and columns swap places.

Mathematics

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \qquad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Python

```
print(A.T)
```

ML Connection

Appears constantly: $X^T X$ in linear regression, $W^T X$ in neural networks, and in the covariance matrix used by PCA.

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7. Dot Product

Real-Life Analogy

Hiring a candidate. You weight Experience (10), Skills (8), Communication (5). The candidate scores 7, 9, 8 respectively. Multiply each weight by the matching score and add them up to get one overall fit score.

Mathematics

$$\mathbf{w}^T \mathbf{x} = (10)(7) + (8)(9) + (5)(8) = 70 + 72 + 40 = 182$$

Python

```
w = np.array([10, 8, 5])
x = np.array([7, 9, 8])
np.dot(w, x)    # 182
```

ML Connection

The dot product is the heart of ML. A single neuron, linear regression, and logistic regression all compute

$$y = \mathbf{w}^T \mathbf{x} + b.$$

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8. Matrix Multiplication

Real-Life Analogy

Two students' marks in Math and Science are combined with subject weightages (Math 0.4, Science 0.6) to produce each student's final score. Matrix multiplication does this for many students and many subjects at once.

Mathematics

$$A = \begin{bmatrix} 80 & 90 \\ 70 & 85 \end{bmatrix}, \quad B = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}, \quad AB = \begin{bmatrix} 86 \\ 79 \end{bmatrix}$$

Rule: each entry of AB is the dot product of a row of A with a column of B. Inner dimensions must match.

Python

```
A @ B
```

ML Connection

Every forward pass of a neural network is a chain of matrix multiplications $WX + b$.

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9. Identity Matrix

Real-Life Analogy

Multiplying any number by 1 leaves it unchanged. The identity matrix I does the same job for matrices.

Mathematics

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad AI = A$$

Python

```
np.eye(3)
```

ML Connection

Used in matrix inversion, ridge regression ($X^TX + \lambda I$), and many optimization steps.

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10. Inverse Matrix

Real-Life Analogy

Division undoes multiplication: $10 \div 2 = 5$. The inverse A^{-1} undoes the matrix A.

Mathematics

$$AA^{-1} = I$$

Python

```
np.linalg.inv(A)
```

ML Connection

The closed-form solution of linear regression uses an inverse:

$$\beta = (X^TX)^{-1}X^Ty.$$

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11. Determinant

Real-Life Analogy

A health check for a matrix: it tells you whether the matrix can be reversed (inverted). A determinant of zero means the matrix is "broken".

Mathematics

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad |A| = (1)(4) - (2)(3) = -2$$

Python

```
np.linalg.det(A)
```

ML Connection

Before computing A^{-1}, the determinant must be non-zero. Also appears in probability density formulas (e.g. multivariate Gaussian).

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12. Rank

Real-Life Analogy

A table with one column `Height (cm)` and another column `Height (cm) again` carries duplicate information. Rank counts how many truly independent columns there are.

Mathematics

rank(A) = number of linearly independent rows (or columns).

Python

```
np.linalg.matrix_rank(A)
```

ML Connection

Detects duplicate features and multicollinearity, both of which break regression models.

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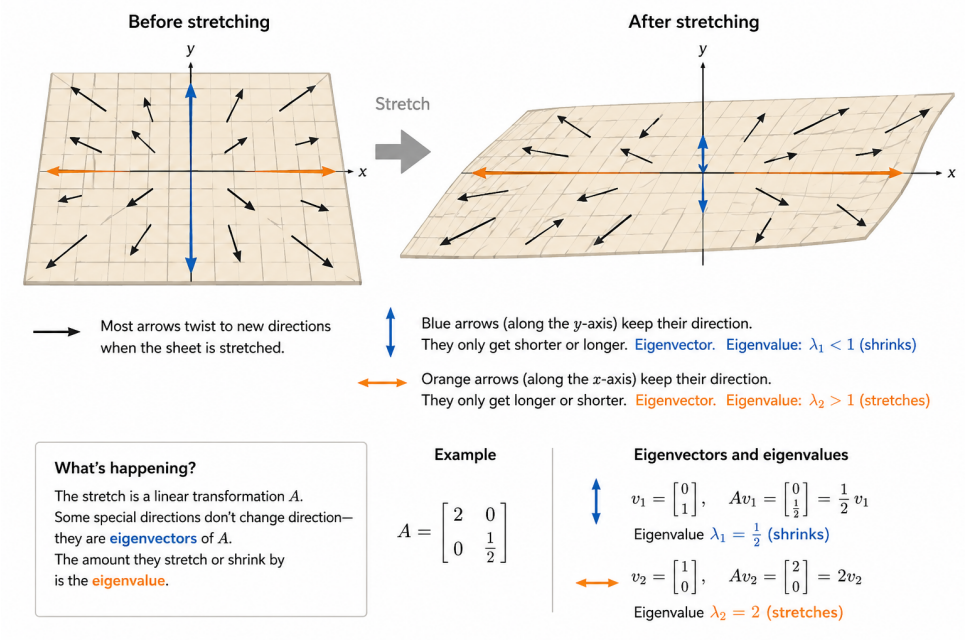
13. Eigenvalues and Eigenvectors

Real-Life Analogy

Stretch a rubber sheet. Most arrows drawn on it twist into new directions. A few special arrows keep pointing the same way and only get longer or shorter. Those arrows are eigenvectors; the stretch factor is the eigenvalue.



Mathematics

$$A\mathbf{v} = \lambda\mathbf{v}$$

where $\mathbf{v}$ is an eigenvector and λ is its eigenvalue.

Python

```
np.linalg.eig(A)
```

ML Connection

The backbone of PCA, face recognition (eigenfaces), image compression, spectral clustering, and PageRank.

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14. Linear Equations

Real-Life Analogy

2 pens + 1 pencil cost Rs. 50; 1 pen + 1 pencil cost Rs. 30. Find the price of each. That is a system of linear equations.

Mathematics

$$\begin{cases} 2x + y = 50 \\ x + y = 30 \end{cases} \iff AX = B$$

Python

```
A = np.array([[2, 1], [1, 1]])
B = np.array([50, 30])
np.linalg.solve(A, B)
```

ML Connection

Linear regression, least squares fitting, and many optimization routines reduce internally to solving $AX = B$.

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15. Principal Component Analysis (PCA)

Real-Life Analogy

A student has marks in Math, Physics, and Chemistry — all highly correlated. Instead of keeping three columns, summarise them as one new column called "Science Performance". PCA does this automatically: it finds the few directions that explain most of the variation in the data.

Mathematics

1. Compute the covariance matrix of the features.
2. Find its eigenvalues and eigenvectors.
3. Keep the eigenvectors with the largest eigenvalues — these are the principal components.

Python

```
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
X_new = pca.fit_transform(X)
```

ML Connection

Used for dimensionality reduction, 2D/3D visualization, noise removal, and faster model training.

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16. Neural Network Application

Real-Life Analogy

A factory line. Inputs (study hours, attendance, assignments) enter, a processing unit weighs and combines them, and an output (pass or fail) comes out the other end. Each neuron is one such processing unit.

Mathematics

A single neuron computes

$$z = \mathbf{w}^T\mathbf{x} + b, \qquad a = \sigma(z),$$

where σ is an activation function (e.g. sigmoid, ReLU).

Python

```
z = np.dot(w, x) + b
a = 1 / (1 + np.exp(-z))    # sigmoid
```

ML Connection

A deep network is just many layers of these neurons — stacks of matrix multiplications, additions, and non-linear activations.

The Hierarchy to Remember

Scalar → Vector → Matrix → Transpose → Dot Product → Matrix Multiplication

→ Linear Equations → Eigenvalues → PCA → Neural Networks → Deep Learning